Quantum cosmology with a geometrical time

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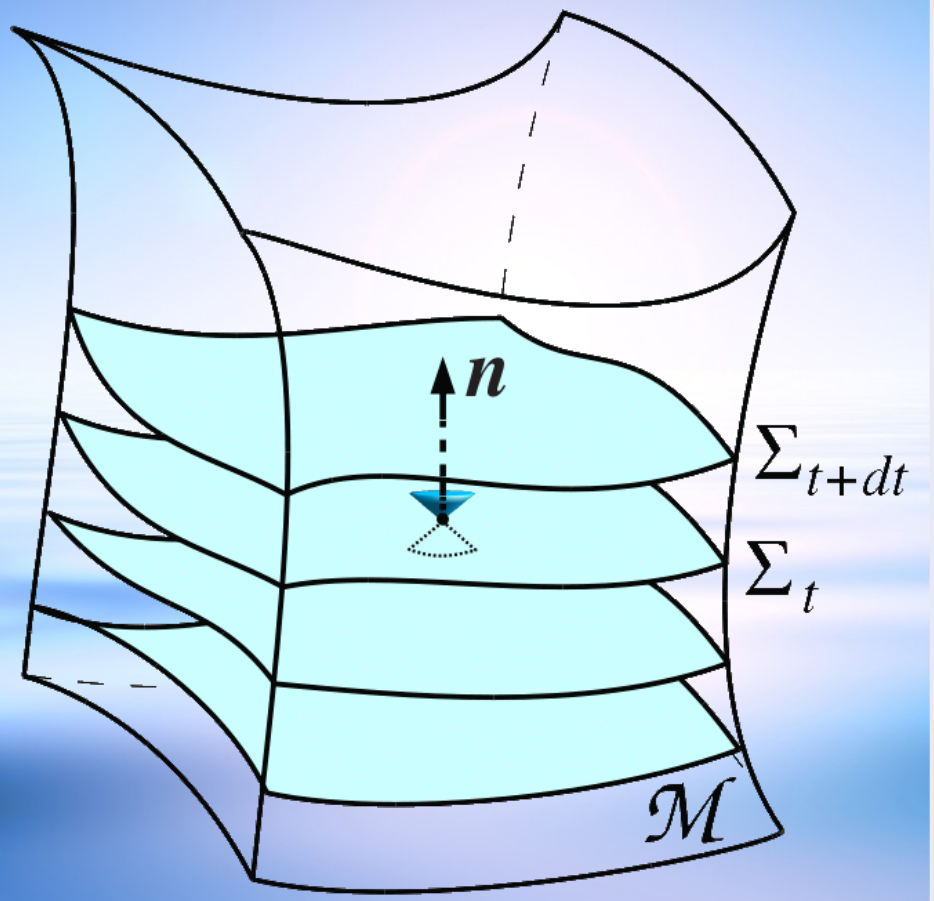
The road to quantum cosmology...

J.J. Halliwell, *Introductory Lectures on Quantum Cosmology* - Quantum Cosmology and Baby Universe

> *"Quantum cosmology is the application of quantum theory to the dynamical systems describing cosmological models."*

– ADM Formalism: A very brief summary:

- Foliation of spacetime on spacelike hypersurfaces, Σ_t;
- Lagrangian function decomposed in terms this foliation;
- Legendre transformation to obtain the super-Hamiltonian, $\mathcal{H}$;
- Once identified the canonical variables in Legendre transformation, proceeds to the formalism of canonical quantization, *à la Dirac*.



Splitting of the spacetime $\mathcal{M}$

A scalar field $t(x^\alpha)$ is defined in the manifold $\mathcal{M}$ such that

$$t = \text{constant}$$

describes a family of spacelike hypersurfaces Σ_t.

This define a unit normal vector $n_\alpha \propto \partial_\alpha t$, which, to ensure proper normalization, it is introduced the *lapse function* N,

$$n_\alpha \doteq -N\partial_\alpha t \;\therefore\; n_0 = -N \;,\; n_i = 0$$

Spacetime decomposition $(3+1)$

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu \equiv -\left(N^2 - N_i N^i\right) dt^2 + 2N_i dx^i dt + h_{ij} dx^i dx^j \qquad (1)$$

where we identify,

$$g_{00} = N^i N_i - N^2 \ , \ g_{0i} = N_i \ , \ g_{ij} = h_{ij}$$

with the contravariant components,

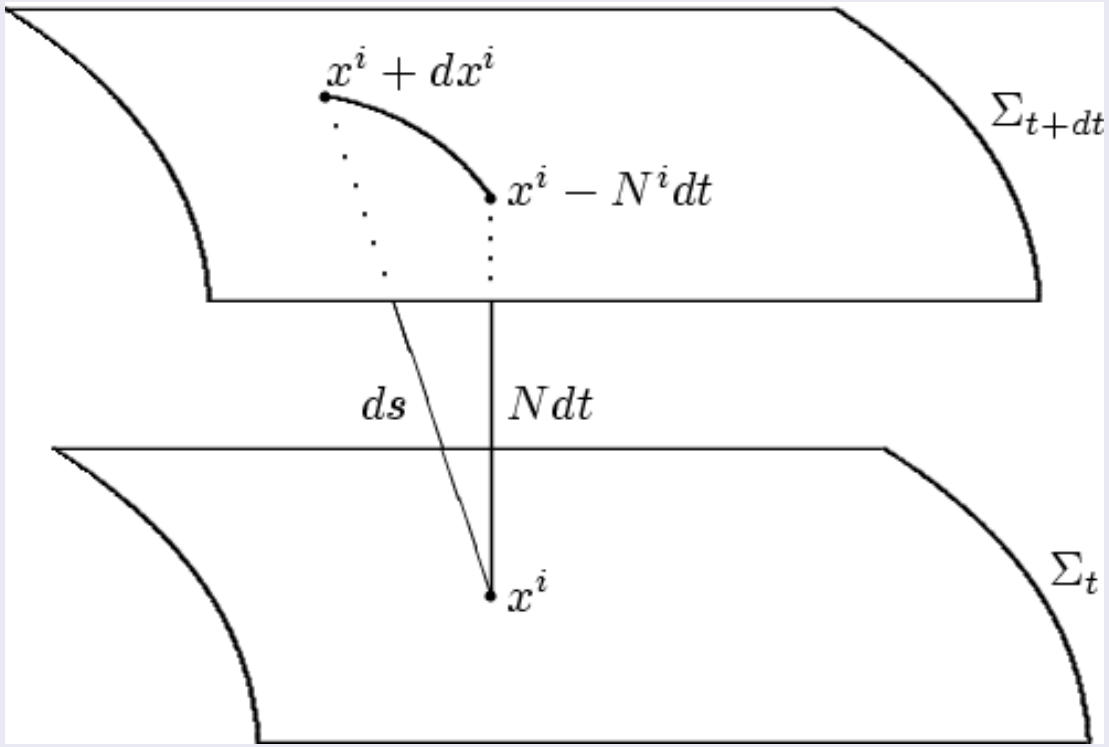
$$g^{00} = -N^{-2} \ , \ g^{0j} = \frac{N^j}{N^2} \ , \ g^{ij} = h^{ij} - \frac{N^i N^j}{N^2}$$

Furthermore, we have the projector tensor $h_{\mu\nu}$ defined as,

$$h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu \implies h_{00} = N^i N_i \ , \ h_{0i} = N_i$$

Therefore, the *lapse function* N and components of *shift vector field* N_i describe how each of the *leaves* Σ_t of the foliation of spacetime are "welded" together.

– links between the leaves –



Einstein-Hilbert Lagrangian in ADM formalism

E-H Action

$$\mathcal{S}_{EH} = \frac{1}{16\pi}\int d^4x\sqrt{-g}\widetilde{R} \tag{2}$$

where $\widetilde{R}$ is the Ricci scalar computed with the Christoffel symbols $\{{}^{\alpha}_{\mu\nu}\}$.

By decomposing $\widetilde{R}$, we get

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$$\widetilde{R} \equiv \widetilde{\mathcal{L}}_{ADMR} + 2\widetilde{\nabla}_{\mu}\left(\widetilde{a}^{\mu} + n^{\mu}\widetilde{K}\right) \tag{3}$$

where $\widetilde{a}^{\mu} \doteq g^{\mu\nu}n^{\beta}\widetilde{\nabla}_{\beta}n_{\nu}$ and $\widetilde{K}$ is the trace of $\widetilde{K}_{\mu\nu} \doteq -h^{\alpha}_{\mu}h^{\beta}_{\nu}\widetilde{\nabla}_{\alpha}n_{\beta}$

Where we are defining

ADM Lagrangian

$$\widetilde{\mathcal{L}}_{ADMR} \doteq {}^{(3)}\widetilde{R} - \widetilde{K}^{\mu}{}_{\nu}\widetilde{K}^{\nu}{}_{\mu} + \widetilde{K}^{2} \tag{4}$$

Therefore, we can write the Eq.(2) as,

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$$\mathcal{S}_{EH} = \frac{1}{16\pi}\int d^4x\sqrt{-g}\widetilde{\mathcal{L}}_{ADMR} + Surface\ terms \tag{5}$$

where,

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$$Surface\ terms \equiv \frac{1}{8\pi}\int d^4x\sqrt{-g}\widetilde{\nabla}_{\mu}\left(\widetilde{a}^{\mu} + n^{\mu}\widetilde{K}\right) \tag{6}$$

To obtain the variation of this action under variations of the metric, we need to vary N, N_i, h_{ij} and their conjugated canonical momenta independently.

But initially our variables are N, N_i, h_{ij} and they appear in the action in different ways:

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- N appears in the Ndt factor of integration and in the factor $1/N$ expanding $\widetilde{K}_{ij}$

$$\sqrt{-g} \equiv N\sqrt{h} \quad ; \quad \widetilde{K}_{ij} = \frac{1}{2N}\left(D_i N_j + D_j N_i - h_{ij,0}\right) \tag{7}$$

- N_i also appears in $\widetilde{K}_{ij}$
- the time derivatives on N and N_i do not appear in the action
- time derivative of h_{ij} appears in $\widetilde{K}_{ij}$

Therefore,

Constraint equations

$$\frac{\partial}{\partial \dot{N}}\left(\sqrt{-g}\widetilde{\mathcal{L}}_{ADMR}\right) = \frac{\partial}{\partial \dot{N}_i}\left(\sqrt{-g}\widetilde{\mathcal{L}}_{ADMR}\right) \equiv 0 \tag{8}$$

While that,

Canonical momentum

$$\Pi^{ij}_{(h)} \doteq \frac{\partial}{\partial \dot{h}_{ij}}\left(\sqrt{-g}\widetilde{\mathcal{L}}_{ADMR}\right) \equiv \frac{\sqrt{h}}{16\pi}\left(\widetilde{K}^{ij} - \widetilde{K}h^{ij}\right) \tag{9}$$

Given the canonical momentum, we can determine the canonical Hamiltonian through the Legendre transformation,

Legendre transformation

$$H_c = \Pi^{ij}_{(h)}\dot{h}_{ij} - \sqrt{-g}\widetilde{\mathcal{L}}_{ADMR} \tag{10}$$

Thus, we are going to have

$$H_c \equiv N\mathcal{H} + N^i\mathcal{H}_i \tag{11}$$

Since N and Ni are playing the role of Lagrange multipliers, so:

Momentum constraint

$$\mathcal{H}_i = -2\widetilde{D}_j\Pi^j_{\ i} \equiv 0 \tag{12}$$

and

Hamiltonian constraint or super-Hamiltonian

$$\mathcal{H} = 16\pi G_{ijkl}\Pi^{ij}\Pi^{kl} - \frac{\sqrt{h}}{16\pi}\,{}^{(3)}\widetilde{R} \equiv 0 \tag{13}$$

where

DeWitt metric

$$G_{ijkl} = \frac{\sqrt{h}}{2}\left(h_{ik}h_{jl} + h_{il}h_{jk} - h_{ij}h_{kl}\right) \tag{14}$$

Superspace and Wheeler-DeWitt equation

In the canonical quantization procedure, the quantum state of the system is represented by a wave functional $\Psi[h_{ij}, \Phi_{matter}]$, a functional on superspace. The DeWitt metric provides a metric on superspace.

J.J. Halliwell, *Introductory Lectures on Quantum Cosmology* - Quantum Cosmology and Baby Universe

> *"**Superspace** is the arena in which the classical dynamics takes place, the space of all three-metrics and matter field configurations on a three-surface."*

Thus, the wave functional is annihilated by the operator version of the classical Hamiltonian constraint. If one makes the canonical Dirac quantization procedure, we are going to get

Wheeler-DeWitt Equation

$$\mathcal{H}\Psi \equiv \left[-G_{ijkl} \frac{\delta}{\delta h_{ij}} \frac{\delta}{\delta h_{kl}} - \sqrt{h}\, {}^{(3)}\widetilde{R} \right] \Psi = 0 \tag{15}$$

The problem of time

We can note that the Wheeler-Dewitt equation lacks an external time parameter. Apart from the observation that we have no coordinate and thus also no time-coordinate dependence in the Wheeler-DeWitt equation, we find that we have no preferred evolution parameter. This is usually called the *problem of time*.

Problem of time

The absence of an external time parameter and the non-uniqueness of internal timelike variables.

We are going to follow the approach of *"Time before quantization - internal time"* to deal this trouble.

Internal Time

M. Bojowald, *Canonical Gravity and applications* - Cosmology, Black Holes and Quantum Gravity

- Coordinate time in general relativity appears as a gauge parameter, and the dynamics is fully constrained. There is no non-trivial Hamiltonian which would take non-zero values on solutions to the constraints and generate any noticeable change with respect to some absolute time.
- If we decide to forgo gauge parameters and gauge-dependent quantities, time in general relativistic systems must be relational: we can notice change not by the motion of objects with respect to an external time parameter, but by the relative change of one object or degree of freedom with respect to another one.
- This viewpoint has been and is still being developed by Bergmann (1961); Rovelli (1991) and Dittrich (2006, 2007).

- The choice of the object whose change is to be described usually depends on the physical questions asked, for instance the growth of the scale factor $A(t)$ in cosmology. Its change, in a gauge-independent manner, must be described with respect to another degree of freedom, such as a matter field ϕ. The choice of this reference degree of freedom, then called internal time, is rather arbitrary.

- Ideally, one would like a quantity that has a monotonic relationship with coordinate time t in a gauge-dependent formulation. Any event characterized by a fixed value of ϕ would then correspond to a unique time slice $t = const$.

- However, finding such a variable is very difficult, except for some simple models; this constitutes the problem of time, reviewed by Kuchar (1992).

Weyl integrable manifold

C. Romero *et al*, *General relativity and Weyl geometry*, Class. Quantum Grav., 29, (2012)

This geometry is characterized by the *nonmetricity condition*

$$\nabla_\alpha g_{\mu\nu} = \phi_\alpha g_{\mu\nu}, \tag{16}$$

where ϕ is a scalar field, which we shall call *Weyl field*, and we define $\phi_\alpha \doteq \partial_\alpha \phi$.

The Eq.(16) is invariant under the simultaneous transformations

$$\bar{g}_{\mu\nu} = e^f g_{\mu\nu} \quad , \quad \bar{\phi} = \phi + f \tag{17}$$

and determines the affine connection (in this case we shall call *Weyl Connection*),

$$\Gamma^\alpha_{\mu\nu} = \left\{{}^{\alpha}_{\mu\nu}\right\} - \frac{1}{2} g^{\alpha\beta} \left(g_{\beta\mu}\phi_\nu + g_{\beta\nu}\phi_\mu - g_{\mu\nu}\phi_\beta \right). \tag{18}$$

Weyl-like theories

Integrable case:

$$\mathcal{S}_G = \int d^4x\sqrt{-g}f(\phi)\left[R + j(\phi)g^{\mu\nu}\phi_\mu\phi_\nu\right] \tag{19}$$

Where R is the *Ricci scalar* computed with the Weyl connection, $f(\phi)$ and $j(\phi)$ are functions that "label" a specific theory.

For example,

WIST - M. Novello *et al*:

$$\mathcal{S}_G = \int d^4x\sqrt{-g}\left(R + \omega\phi_\mu\phi^\mu\right) \tag{20}$$

Thus,

$$f(\phi) \equiv 1 \quad , \quad j(\phi) \equiv \omega$$

Opening the terms

We can write Ricci scalar in terms of riemannian quantities and terms of ϕ.

Ricci tensor in riemannian terms and ϕ-terms:

$$R_{\mu\nu} \equiv \widetilde{R}_{\mu\nu} - \frac{3}{2}\widetilde{\nabla}_\nu \phi_\mu + \frac{1}{2}\widetilde{\nabla}_\mu \phi_\nu - \frac{1}{2}\phi_\mu \phi_\nu + \frac{1}{2} g_{\mu\nu}\left(\phi^\alpha \phi_\alpha - \widetilde{\nabla}_\alpha \phi^\alpha\right) \quad (21)$$

and, therefore,

Ricci scalar in riemannian terms and ϕ-terms:

$$R = \widetilde{R} - 3\widetilde{\Box}\phi + \frac{3}{2} g^{\mu\nu}\phi_\mu \phi_\nu, \quad (22)$$

This way, the Eq.(19) is given as follow:

Gravitational action:

$$\mathcal{S}_G = \int d^4x\sqrt{-g}f(\phi)\left[\widetilde{\mathcal{L}}_{ADMR} + \left(\frac{3}{2} + j(\phi)\right)g^{\mu\nu}\phi_\mu\phi_\nu\right] - 2D_{iv} \quad (23)$$

We define,

Terms with divergences:

$$D_{iv} \doteq \int d^4x\sqrt{-g}f(\phi)\left[\widetilde{\Box}\phi + \widetilde{\nabla}_\alpha\left(\xi n^\alpha\right)\right] \quad (24)$$

where

$$\xi \doteq g^{\mu\nu}n_\mu\phi_\nu \equiv \frac{\dot{\phi}}{N^2}$$

General expression in Weyl integrable

By dealing with D_{iv}, to put away the real surface terms, we are going to get:

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$$\mathcal{S}_G = \int d^4x\sqrt{-g}f(\phi)\left[\widetilde{\mathcal{L}}_{ADMR} + \lambda\phi_\mu\phi^\mu\right] + 2F_\phi + \textit{Term. Superf.} \quad (25)$$

where we define,

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$$F_\phi \doteq \int d^4x\sqrt{-g}\left[f'\left(\frac{\dot{\phi}}{N^2} + \phi\widetilde{\Box}\phi\right) + \phi f''\phi_\mu\phi^\mu\right] \quad (26)$$

and also,

$$\lambda \doteq \frac{3}{2} + j(\phi) \quad , \quad f' \doteq \frac{\partial f}{\partial\phi} \implies f'' \doteq \frac{\partial^2 f}{\partial\phi^2}$$

Weyl field as internal time

We have obtained so that the field Weyl is identified as a canonical variable, since there will be a conjugated canonical momentum to it. The functional form of this momentum depends on $f(\phi)$ and $j(\phi)$.

For the case of WIST, where $f(\phi) = \text{const.}$ and $\lambda = \omega$, we have:

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$$\mathcal{S}_G = \int d^4x\sqrt{-g}\widetilde{\mathcal{L}}_{ADMR} + \int d^4x\sqrt{-g}\omega g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi \tag{27}$$

Namely, the expression for the action is identical (with $\omega = 1/2$) to the case of quantum cosmology applied to General Relativity minimally coupled to a massless scalar field.

¡Muchas gracias!

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